GPSC CBRT · HOMOEOPATHIC PHYSICIAN · CORE

# Topic 10: Community Medicine, Forensic Medicine & Ethics

Complete notes with pictures, memory hooks, rapid revision and practice MCQs.

# Topic 10: Community Medicine, Forensic Medicine & Ethics

## How to use this doc

- This is study material for **Core Topic 10** of the GPSC Homoeopathic Physician CBRT (Advt 06/2026, exam **1 November 2026**).
- Core carries 25 of the 60 marks across Topics 1–10, so expect **2–3 questions** from this topic.
- They are usually one-line facts: a definition, a formula, a colour code, a section of an Act, or a classical sign.

### What it covers

- 10.1 Epidemiology, levels of prevention, screening, and National Health Programmes, including vaccines.
- 10.2 Demography, vital statistics, basic biostatistics, water, sanitation and biomedical waste.
- 10.3 Forensic medicine and toxicology: death and post-mortem changes, injuries, asphyxia, common poisons.
- 10.4 Medical jurisprudence and ethics for a homoeopathic physician: NCH Act 2020, professional conduct, consent, negligence, records.

### Standard books followed

- K. Park, *Park's Textbook of Preventive and Social Medicine*.
- B.K. Mahajan, *Methods in Biostatistics*.
- K.S.N. Reddy, *The Essentials of Forensic Medicine and Toxicology*.
- C.K. Parikh, *Textbook of Medical Jurisprudence, Forensic Medicine and Toxicology*.
- Modi's *Medical Jurisprudence and Toxicology*.
- The National Commission for Homoeopathy Act, 2020, and the professional conduct regulations for homoeopathic practitioners.

**Scope note:** conventional (allopathic) drug treatment is left out. Vaccines and programme schedules are kept because they are public-health facts. Antidotes appear only as forensic toxicology facts that examiners ask.

**How to revise:** read each section once, look at the diagram, then do the Rapid revision list and the 20 MCQs. Mark any wrong answer and re-read only that paragraph.

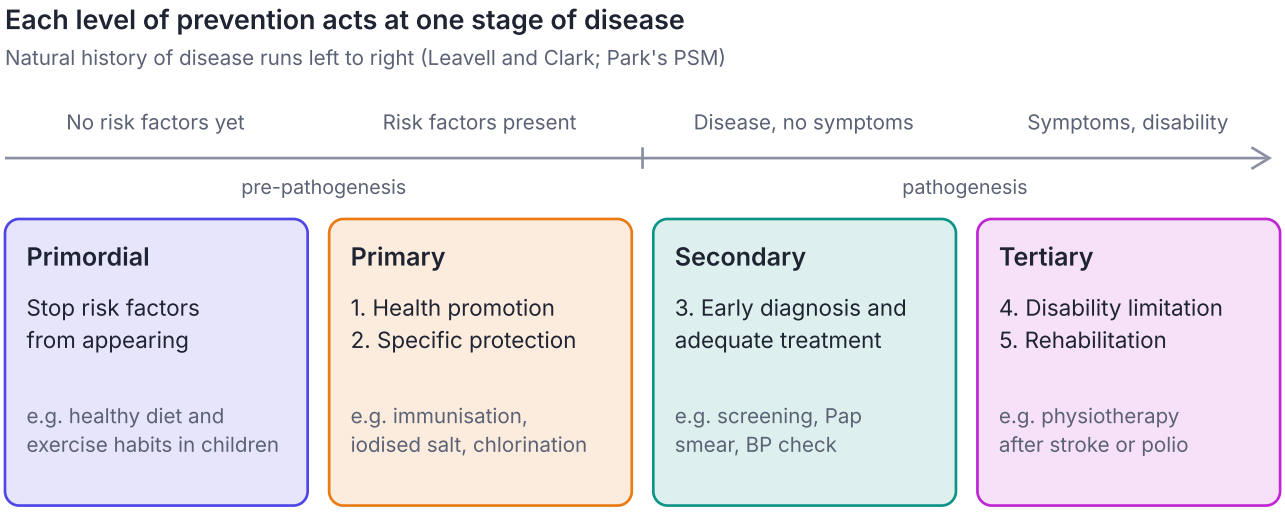
## 10.1 Epidemiology, prevention and National Health Programmes

### Health and disease

- **WHO definition of health (1948):** "a state of complete physical, mental and social well-being and not merely an absence of disease or infirmity". Spiritual health is a proposed fourth dimension.
- **Epidemiological triad:** agent, host and environment. For non-communicable diseases a **web of causation** (multiple factors) is used instead.
- **Iceberg phenomenon:** the visible tip is clinical cases; the submerged part is sub-clinical, undiagnosed and carrier cases. Hypertension, diabetes and malnutrition show a large hidden part.
- **Epidemiology** is the study of the distribution and determinants of health states in populations and its application to their control (Last).

### Levels of prevention

Prevention is matched to the stage of disease. Leavell and Clark gave **five modes of intervention**: health promotion, specific protection, early diagnosis and treatment, disability limitation, and rehabilitation.




Levels of prevention · five modes of intervention across the natural history of disease

Read the chart left to right: primordial prevention acts before any risk factor exists, and tertiary prevention acts once disease has caused disability.

### Measuring disease

- **Incidence:** new cases in a period ÷ population at risk. It measures risk and is used for acute diseases.
- **Prevalence:** all existing cases (old + new) at a time ÷ total population. It is useful for chronic diseases and planning services.
- **Prevalence = Incidence × Mean duration** (P = I × D).
- **Attack rate:** an incidence rate used during an epidemic. **Secondary attack rate** = cases among contacts ÷ susceptible contacts; it measures infectiousness.
- **Incubation period:** time from infection to first symptom. **Serial interval:** gap between onset of primary and secondary cases.
- **Herd immunity:** a high proportion of immune people protects the unimmunised; it underlies mass immunisation.

## Epidemics

- **Common-source, single exposure:** cases rise and fall within one incubation period (e.g. food poisoning).
- **Propagated (person to person):** gradual rise, several waves (e.g. hepatitis A, measles).
- **Endemic:** constant presence in an area. **Pandemic:** spread over many countries.
- **Sporadic:** scattered, irregular cases.

## Study designs (a frequent question)

| Design | Starts with | Gives | Example use |
|---|---|---|---|
| Cross-sectional | A population at one time | Prevalence | Survey of anaemia |
| Case-control | Cases and non-cases, looks back to exposure | **Odds ratio** | Rare disease, quick, cheap |
| Cohort | Exposed and non-exposed, follows forward | **Incidence, relative risk, attributable risk** | Smoking and lung cancer (Doll and Hill) |
| Randomised controlled trial | Random allocation to groups | Efficacy of an intervention | Gold standard for therapy |

**Blinding:** single (subject unaware), double (subject and investigator unaware), triple (also the analyst).

## Screening

Use the 2×2 table: TP, FP, FN, TN.

- **Sensitivity** = TP ÷ (TP + FN): ability to detect true cases (true positive rate).
- **Specificity** = TN ÷ (TN + FP): ability to identify true non-cases.
- **Positive predictive value** = TP ÷ (TP + FP). It **rises with prevalence**.
- **Negative predictive value** = TN ÷ (TN + FN).
- A screening test should be sensitive; a confirmatory (diagnostic) test should be specific.
- Wilson and Jungner (WHO, 1968) gave the criteria for a screening programme.

## Immunisation and the cold chain

- **Active immunity:** from infection or vaccine; slow but long-lasting. **Passive:** ready-made antibodies (e.g. immunoglobulin, maternal antibodies); immediate but short.
- **Cold chain:** vaccines are kept at **+2 to +8 °C** in an ice-lined refrigerator (ILR) at the health facility.
- **Most heat-sensitive vaccine:** OPV. **Freeze-sensitive vaccines** (never freeze): Hepatitis B, pentavalent, DPT, Td, IPV.
- **Shake test** detects a frozen vaccine. **Vaccine vial monitor (VVM):** discard when the inner square matches or is darker than the outer circle.
- The full Universal Immunisation Programme schedule is in the Topic 8 doc.

## Genus epidemicus (homoeopathic public health)

- In an epidemic, symptoms of several patients are pooled to form one picture, and the remedy matching it is the **genus epidemicus** (Organon §100–102).
- Hahnemann used **Belladonna** as a prophylactic in **scarlet fever** (1799). This is the classic historical example.

## National Health Programmes

| Programme | Key facts |
|---|---|
| National TB Elimination Programme (NTEP) | Renamed from RNTCP in 2020; aim to end TB ahead of the SDG 2030 target; Nikshay portal; DOTS strategy |
| National Leprosy Eradication Programme (NLEP) | Multi-drug therapy free; Sparsh awareness campaign on 30 January |
| National Centre for Vector Borne Diseases Control (NCVBDC) | Malaria, dengue, chikungunya, filaria, kala-azar, Japanese encephalitis |
| National AIDS Control Programme (NACP) | 95-95-95 target; ICTC for counselling and testing |

| Programme | Key facts |
|---|---|
| NP-NCD (formerly NPCDCS) | Screening of hypertension, diabetes, common cancers at 30+ years |
| National Programme for Control of Blindness and Visual Impairment | Cataract is the leading cause of blindness |
| National Mental Health Programme | District Mental Health Programme; Tele-MANAS helpline 14416 |
| Pulse Polio, Mission Indradhanush (2014) | India declared polio-free in 2014; Indradhanush covers missed children |
| Rashtriya Bal Swasthya Karyakram (RBSK) | Screens 0–18 years for the 4 Ds: defects at birth, diseases, deficiencies, developmental delays |
| Janani Suraksha Yojana (2005), JSSK (2011) | Cash for institutional delivery; free delivery and care for mother and sick newborn |
| Ayushman Bharat (2018) | Ayushman Arogya Mandir (health and wellness centres) and PM-JAY cover of ₹5 lakh per family per year |

### Health care delivery norms

| Facility | Plain area population | Hilly or tribal population |
|---|---|---|
| Sub-centre | 5,000 | 3,000 |
| Primary Health Centre | 30,000 | 20,000 |
| Community Health Centre (30 beds, 4 specialists) | 1,20,000 | 80,000 |

One **ASHA** serves about 1,000 population. The National Rural Health Mission began in 2005 and became part of the National Health Mission in 2013. **Bhore Committee (1946)** laid the foundation of primary health care in India; the **Alma-Ata Declaration (1978)** set "Health for All".

**Memory hook:** prevention runs **Promote, Protect** (primary), **Diagnose early** (secondary), **Limit disability, Rehabilitate** (tertiary). For screening tests: **SnNout** (a **S**ensitive test, when **N**egative, rules **out**) and **SpPin** (a **S**pecific test, when **P**ositive, rules **in**).

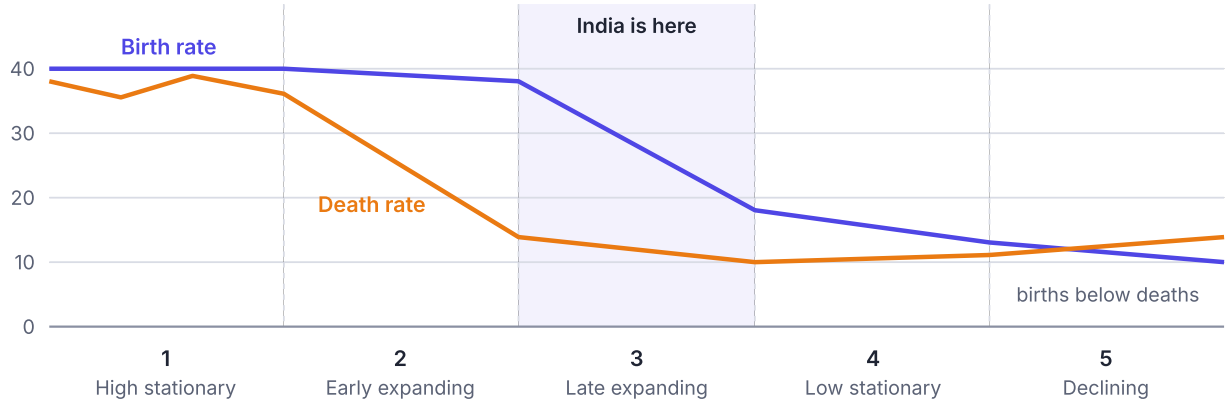
## 10.2 Demography, biostatistics and environmental health

### Demography

Demography studies the size, composition and distribution of a population. Its three processes are **fertility, mortality and migration**.

**Demographic cycle: the gap between birth and death rates drives growth**

Schematic, rates per 1,000 population; India is in the late expanding stage




Stage 2 has the fastest growth: deaths fall while births stay high.

Demographic cycle · five stages of birth and death rates

The population grows fastest when the death rate has fallen but the birth rate is still high; India's birth rate is now falling, which places it in stage 3.

- **Census 2011 (latest completed census):** population 121.09 crore; sex ratio 943 females per 1,000 males; child sex ratio (0–6 years) 919; literacy 74.04%; density 382 per sq km; decadal growth 17.7%.
- Gujarat's sex ratio was 919.
- The census is conducted under the **Census Act, 1948**.

## Vital statistics: definitions and denominators

| Indicator | Formula (denominator is the exam trap) |
|---|---|
| Crude birth rate | Live births in a year per **1,000 mid-year population** |
| Crude death rate | Deaths in a year per 1,000 mid-year population |
| Infant mortality rate (IMR) | Deaths under 1 year per **1,000 live births** |
| Neonatal mortality rate | Deaths in the first 28 days per 1,000 live births |
| Perinatal mortality rate | Still births (28 weeks+) and deaths in the first 7 days per 1,000 **live births + still births** |
| Maternal mortality ratio (MMR) | Maternal deaths per **1,00,000 live births** |

| Indicator | Formula (denominator is the exam trap) |
|---|---|
| General fertility rate | Live births per 1,000 women aged 15–49 |
| Total fertility rate (TFR) | Average children a woman would have in her reproductive life; India about **2.0** (NFHS-5) |
| Net reproduction rate (NRR) | Daughters per woman surviving to mothers' age; **NRR = 1** equals replacement level (TFR about 2.1) |
| Dependency ratio | (0–14 + 65 and over) ÷ 15–64 population |

IMR is the best single indicator of a community's health status. Registration of births and deaths falls under the **Registration of Births and Deaths Act, 1969**: births within 21 days.

## Biostatistics essentials

- **Data types:** qualitative (sex, blood group) versus quantitative; quantitative is discrete (number of children) or continuous (height, Hb).
- **Mean** is affected by extreme values; **median** is the middle value and is best for skewed data; **mode** is the most frequent value.
- **Measures of dispersion:** range, mean deviation, **standard deviation** (most used), variance ($SD^2$), coefficient of variation = SD ÷ mean × 100.
- **Normal curve:** bell-shaped, mean = median = mode. Mean ± 1 SD covers **68%**, ± 2 SD covers **95%**, ± 3 SD covers **99.7%** of values.
- **Standard error of mean** = SD ÷ √n.
- **Tests:** chi-square for proportions (qualitative data); t-test for two means (small samples); z-test for large samples; ANOVA for three or more means.
- **$p < 0.05$** means the result is statistically significant. A **type I error** (α) is rejecting a true null hypothesis; a **type II error** (β) is accepting a false one.
- **Correlation coefficient (r)** ranges from −1 to +1; it is shown on a **scatter diagram**.
- **Diagrams:** bar chart and pie chart for qualitative data; **histogram** and frequency polygon for continuous data; **ogive** for cumulative frequency; line diagram for a trend over time.

## Water

- **Safe water:** free from pathogens and harmful chemicals, pleasant to drink, usable for domestic purposes. The daily need is about 150–200 litres per person in cities.
- **Purification on a large scale:** storage → filtration → chlorination.

- **Slow sand filter:** the **vital layer (Schmutzdecke)** of algae and bacteria on the sand surface does the work. **Rapid sand filter:** needs prior coagulation with alum; cleaned by back-washing.
- **Chlorination:** the free residual chlorine must be **0.5 mg/L after 1 hour of contact**. **Break-point chlorination** is the point after which added chlorine appears as free residual chlorine.
- **Horrock's apparatus** finds the dose of bleaching powder for well water. Bleaching powder contains about **33% available chlorine**.
- **Fluoride:** desirable level 0.5–0.8 mg/L; above 1.5 mg/L causes fluorosis; correction is by the **Nalgonda technique**.
- **Hardness:** temporary hardness (bicarbonates) is removed by boiling; permanent hardness (sulphates, chlorides) by the base-exchange process.

### Sanitation: excreta and refuse disposal

Faecal-oral diseases (typhoid, cholera, hepatitis A, polio, worms, diarrhoea) spread along the **F-diagram**: Faeces → Fingers, Flies, Fields, Fluids → Food. A **sanitation barrier**, the safe disposal of excreta, breaks every one of these routes at the start.

| Method | Key facts |
|---|---|
| Sanitary latrine | Must not pollute the soil or water, and must keep excreta away from flies and animals; sited at least 15 m (about 50 feet) from a water source |
| Water-seal latrine | A trap of water seals the pan from the pit, keeping out flies and smell; the RCA (Research-cum-Action) type is suited to rural India |
| Pit and bore-hole latrines | Simple dry pits for places without a water supply |
| Septic tank | For houses without sewers: solids settle and are digested anaerobically; the effluent drains to a soak pit |
| Sewerage | Water carriage of sewage to a treatment plant (towns and cities) |
| Refuse (solid waste) | Disposed of by composting, sanitary landfill or incineration |

**Swachh Bharat Mission** (launched 2 October 2014) aims at an open-defecation-free (ODF) India, with household toilets and solid-waste management.

**Memory hook:** the five Fs (faeces, fingers, flies, fields, fluids) lead to food; a latrine 15 metres from the well is the first barrier.

## Air, noise and housing

- Indicators of air pollution: sulphur dioxide, smoke, grit and dust. Ventilation standard: about 3,000 cubic feet of fresh air per person per hour.
- Noise above **85 dB** risks hearing loss over long exposure.

## Biomedical waste: colour codes (Bio-Medical Waste Management Rules, 2016)

| Bag or container | What goes in | Final treatment |
|---|---|---|
| **Yellow** | Human anatomical waste, soiled dressings and cotton, expired or discarded medicines, chemical waste, microbiology cultures | Incineration or deep burial |
| **Red** | Contaminated recyclable plastics: tubing, bottles, IV sets, catheters, syringes **without needles**, gloves | Autoclave or microwave, shred, then recycle |
| **White (translucent), puncture-proof** | Sharps: needles, syringes with fixed needles, scalpels, blades | Autoclave, shred, sharps pit or encapsulation |
| **Blue (cardboard box)** | Glassware (broken or discarded) and metallic body implants | Disinfect or autoclave, then recycle |

## Biomedical waste: which colour?

Segregate at the source, by colour



### Yellow

bag

**WHAT GOES IN**

**Anatomical waste**
**Soiled dressings, cotton**
**Expired medicines**
**Chemical waste**
**Microbiology cultures**

**TREATMENT**

Incineration or deep burial



### Red

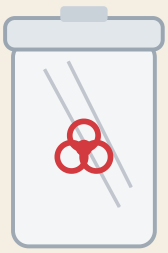
bag

**WHAT GOES IN**

**Recyclable plastics: tubing, bottles, IV sets, catheters, gloves, syringes WITHOUT needles**

**TREATMENT**

Autoclave or microwave, shred, recycle



### White

puncture-proof, translucent

**WHAT GOES IN**

**Sharps: needles, syringes with fixed needles, scalpels, blades**

**TREATMENT**

Autoclave, shred, sharps pit or encapsulation



### Blue

cardboard box

**WHAT GOES IN**

**Glassware (broken or discarded)**
**Metal body implants**

**TREATMENT**

Disinfect or autoclave, then recycle

Bio-Medical Waste Management Rules, 2016 · treat within 48 hours · hook: Yellow body, Red plastic, White sharps, Blue glass

Biomedical waste · Yellow body, Red plastic, White sharps, Blue glass

Waste must be treated and disposed of within **48 hours**. Bags must not be chlorinated plastic (no PVC).

**Memory hook:** waste bags by colour: **Yellow** for the body (anatomical and soiled waste), **Red** for recyclable plastic, **White** for sharps, **Blue** for glass. Census 2011 in two numbers: **943** women per 1,000 men and **74.04%** literacy.

## 10.3 Forensic medicine and toxicology

**New criminal laws:** from **1 July 2024** the IPC, CrPC and Indian Evidence Act were replaced by the **Bharatiya Nyaya Sanhita (BNS)**, **Bharatiya Nagarik Suraksha Sanhita (BNSS)** and **Bharatiya Sakshya Adhiniyam (BSA)**. Learn both numbers, as questions may use either.

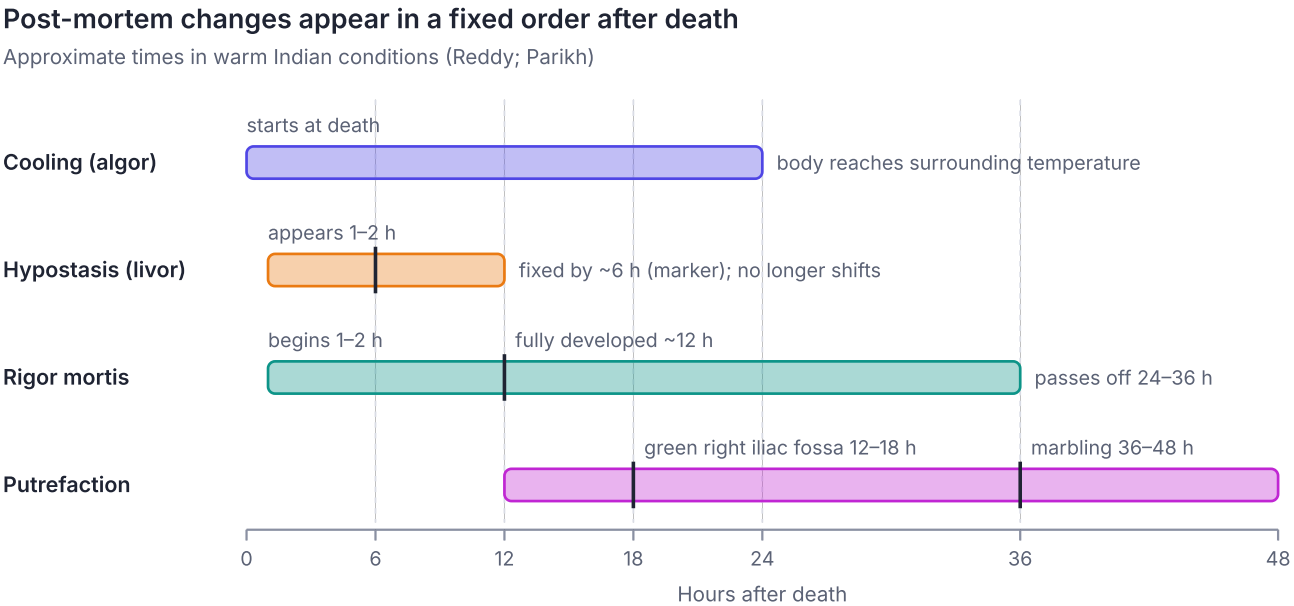
| Subject | Old section | New section (2023 Acts) |
|---|---|---|
| Murder (punishment) | IPC 302 | BNS 103 |

| Subject | Old section | New section (2023 Acts) |
|---|---|---|
| Grievous hurt (definition) | IPC 320 | BNS 116 (severe pain period cut from 20 to **15 days**) |
| Police inquest | CrPC 174 | BNSS 194 |
| Magistrate inquest (custodial death) | CrPC 176 | BNSS 196 |
| Dying declaration | Evidence Act 32 | BSA 26 |

## Death and its signs

- **Somatic death:** permanent stop of heart, lungs and brain function. **Molecular (cellular) death** follows over the next hours.
- **Brain-stem death** is recognised for organ donation under the Transplantation of Human Organs and Tissues Act, 1994.
- **Suspended animation:** apparent death (e.g. drowning, electrocution, newborns); resuscitation must be attempted.

## Post-mortem changes




Post-mortem changes · approximate timeline used to estimate time since death

Estimate the time since death by reading several changes together; no single sign is reliable alone.

- **Hypostasis (livor mortis):** blood settles in dependent parts; spared at pressure points (contact flattening). It is **cherry-red in CO and cyanide** poisoning and **chocolate (brown)** in nitrite or potassium chlorate poisoning.

- **Rigor mortis:** follows **Nysten's rule**: eyelids → face → neck → trunk → upper limbs → lower limbs, and passes off in the same order. The heart (myocardium) stiffens first of all.
- **Cadaveric spasm (instantaneous rigor):** stiffening at the moment of death in a tense person, e.g. weeds clutched in drowning or a weapon gripped in suicide. It cannot be faked and is strong evidence of what happened at death.
- **Heat stiffening** (burns, above 65 °C) and **cold stiffening** are different from rigor.
- **Putrefaction:** the first external sign is **greenish discolouration over the right iliac fossa** (caecum lies close to skin). **Casper's dictum:** decay in air, water and earth is in the ratio `1:2:8`.
- **Adipocere (saponification):** body fat turns soapy in warm, moist conditions (water, damp graves). **Mummification:** drying in hot, dry air; features are preserved for identification.

## Identification

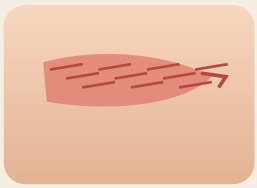
- **Locard's exchange principle:** every contact leaves a trace.
- **Fingerprints (dactylography):** Galton's system; **loops** are the most common pattern; no two persons share fingerprints.
- **Age:** from teeth (Gustafson's method) and ossification of bones; the third molar erupts at 17–25 years.
- **Sex:** pelvis is the best bone, then skull.
- **DNA profiling** is the most accurate method today.

## Mechanical injuries

| Injury | Key forensic feature |
|---|---|
| Abrasion | Only the epidermis is lost; shows the direction of force and the pattern of the object; heals without scar |
| Contusion (bruise) | Bleeding under intact skin; colour changes with age: red → blue (1–3 days) → brown → **green (5–6 days)** → **yellow (7–12 days)** → normal (about 2 weeks) |
| Laceration | Tearing by a blunt force; ragged margins, **tissue bridges** in the wound, bruised edges |
| Incised wound | Sharp-edged weapon; clean cut, longer than deep; **tailing** shows the end of the cut. **Hesitation cuts** suggest suicide; **defence wounds** on palms and forearms suggest homicide |
| Stab wound | Depth greater than length; shape tells single- or double-edged blade |
| Firearm wound | Entry: inverted edges, abrasion collar, burning and tattooing at close range. Exit: everted, larger, no collar |

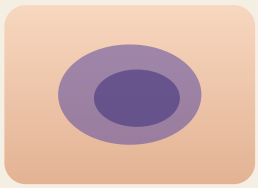
## Mechanical injuries: read the wound

What each wound looks like, and how a bruise ages



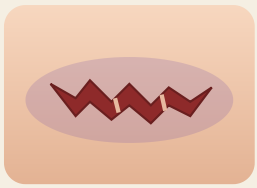
**Abrasion**

Only the epidermis is lost; shows direction of force; heals without a scar



**Contusion (bruise)**

Bleeding under intact skin; its colour dates the injury (strip below)



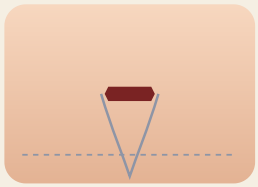
**Laceration**

Blunt force tears: ragged margins, tissue bridges, bruised edges



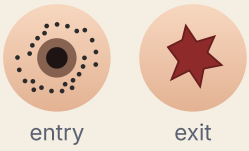

**Incised wound**

Sharp edge: clean cut, longer than deep; the tail shows where it ended



**Stab wound**

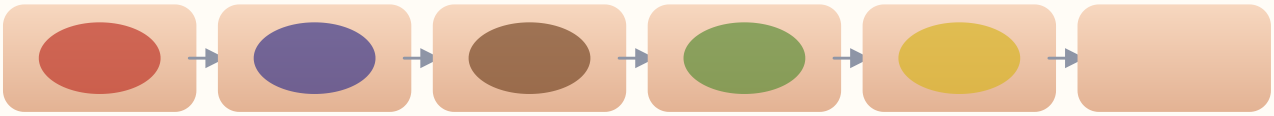
Depth greater than length; shape tells a single- or double-edged blade




**Firearm wound**

Entry: small, inverted, collar, tattooing at close range. Exit: larger, everted, no collar

### A bruise changes colour as it ages



| Red | Blue | Brown | Green | Yellow | Normal |
|---|---|---|---|---|---|
| fresh | 1–3 days | | 5–6 days | 7–12 days | ≈2 weeks |

Mechanical injuries · what each wound looks like, and how a bruise ages

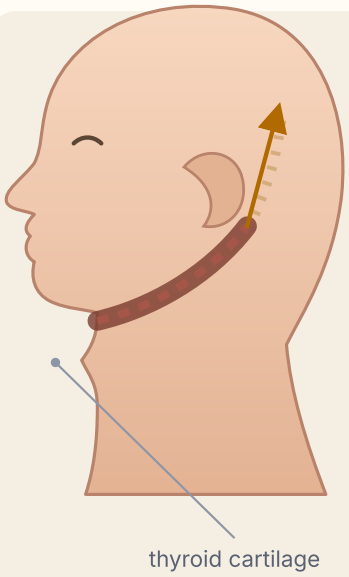
## Asphyxial deaths

| Feature | Hanging | Ligature strangulation |
|---|---|---|
| Usual manner | **Suicidal** | **Homicidal** |
| Ligature mark position | High, **above the thyroid cartilage** | Low, **at or below** the thyroid cartilage |
| Direction | **Oblique**, rising to the knot | **Horizontal**, transverse |
| Continuity | **Incomplete** (gap at knot) | **Complete**, encircles the neck |

| Feature | Hanging | Ligature strangulation |
|---|---|---|
| Face | Pale | Congested, petechiae |
| Salivary dribbling | Present; a sign of hanging while alive | Absent |

## Hanging vs ligature strangulation

Read the ligature mark: direction, level and continuity




### Hanging

DIRECTION
Oblique, rising to the knot

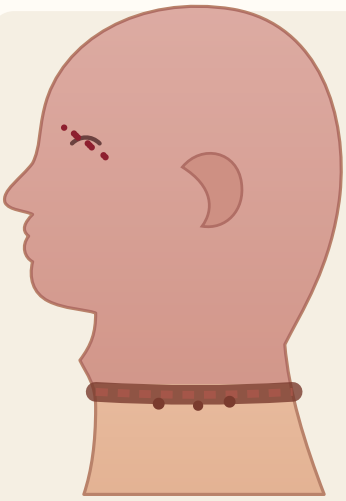
POSITION
High, above the thyroid cartilage

CONTINUITY
Incomplete (gap at the knot)

FACE
Pale

SALIVA DRIBBLING
Present: hanged while alive

USUAL MANNER
Suicidal



### Ligature strangulation

DIRECTION
Horizontal, transverse

POSITION
Low, at or below thyroid cartilage

CONTINUITY
Complete, encircles the neck

FACE
Congested, petechiae

SALIVA DRIBBLING
Absent

USUAL MANNER
Homicidal

Throttling (manual strangulation): finger-nail marks and bruises; hyoid fracture is commonest here.

Hanging vs ligature strangulation · direction, level and continuity of the mark

- **Throttling (manual strangulation):** finger-nail marks and bruises on the neck; the **hyoid bone fracture** is commonest here.
- **Drowning:** fine, persistent **froth at mouth and nostrils**; weeds in the hand (cadaveric spasm); **diatoms** in bone marrow confirm death by drowning.
- **Tardieu spots:** petechial haemorrhages under pleura and pericardium in asphyxia.

### Poisons: classical signs examiners ask

Many of these poisons are also homoeopathic polychrests; their toxic picture forms part of the proving.

| Poison | Classical forensic facts |
|---|---|
| Organophosphate insecticides | Commonest poisoning in rural India; **pin-point pupils**, salivation, sweating, fasciculations; kerosene-like smell; low cholinesterase; atropine is the standard antidote |
| Datura (*Datura stramonium*) | "**Dry as a bone, red as a beet, blind as a bat, mad as a hatter, hot as a hare**"; dilated pupils; seeds resemble chilli seeds; used by road poisoners |
| Arsenic | Rice-water stools; **Mees' lines** on nails; raindrop pigmentation; detected in hair and nails long after; delays putrefaction; **Marsh and Reinsch tests** |
| Lead | **Burtonian blue line** on gums; basophilic stippling of red cells; wrist drop; colic |
| Mercury | Acrodynia (pink disease); Minamata disease from methyl mercury |
| Cyanide | Fastest-acting poison; **bitter-almond smell**; bright red hypostasis |
| Carbon monoxide | **Cherry-red** colour of blood and hypostasis |
| Opium (morphine) | Pin-point pupils, deep coma, slow breathing |
| Aconite | Most poisonous plant in India; tingling and numbness of mouth and limbs |
| Strychnine (*Nux vomica* seed) | Convulsions with **opisthotonus**, risus sardonicus; mind clear between fits |
| Oleander (*Nerium*, yellow oleander) | Cardiac poison; irregular slow pulse |
| *Abrus precatorius* (ratti) | Seeds made into **sui** (needles) to kill cattle |
| *Calotropis* (madar) | Milky juice used as an abortifacient and for infanticide |

### Snake bite

- **India's "big four":** common cobra, common krait, Russell's viper, saw-scaled viper.
- **Elapids (cobra, krait)** are **neurotoxic**: ptosis, difficulty swallowing, respiratory paralysis. Krait bites at night and is often painless.
- **Vipers** are **haemotoxic**: local swelling and bleeding. The **20-minute whole blood clotting test** shows non-clotting blood.
- First aid: reassure, immobilise the limb, no cutting or sucking, and transfer quickly for **polyvalent anti-snake venom**, which covers the big four.

**Memory hook:** the new laws keep their initials: **N**yaya replaces the penal code, **N**agarik **S**uraksha replaces the CrPC, **S**akshya replaces the Evidence Act (BNS, BNSS, BSA from 1 July 2024). A **hanging** mark climbs **obliquely above** the thyroid and breaks; a **strangulation** mark runs **flat and complete** at or below it.

## 10.4 Medical jurisprudence and ethics

### Regulating homoeopathy in India

| Milestone | Fact |
|---|---|
| Homoeopathy Central Council Act, 1973 | Set up the Central Council of Homoeopathy (CCH) to keep the Central Register and set education standards |
| Department of ISM&H (1995) → Department of AYUSH (2003) → **Ministry of AYUSH (November 2014)** | Government body for Ayurveda, Yoga, Unani, Siddha, Sowa-Rigpa and Homoeopathy |
| **National Commission for Homoeopathy (NCH) Act, 2020** | Replaced the 1973 Act; the Commission was constituted in **July 2021** |
| Three autonomous boards under NCH | **Homoeopathy Education Board**; **Medical Assessment and Rating Board for Homoeopathy**; **Board of Ethics and Registration for Homoeopathy** |
| National Register | Kept by the Board of Ethics and Registration; only registered practitioners may practise homoeopathy |
| Central Council for Research in Homoeopathy (CCRH), 1978 | Apex research body under the Ministry of AYUSH |

**State level:** the Gujarat Board of Homoeopathic System of Medicine keeps the State Register for practitioners in Gujarat.

### Cardinal principles of ethics

- **Autonomy** (respect the patient's choice), **beneficence** (do good), **non-maleficence** (do no harm, *primum non nocere*), **justice** (fairness in care).
- **International codes:** Hippocratic Oath; **Declaration of Geneva (1948)**, the modern physician's pledge; **Helsinki (1964)**, research on human subjects; **Tokyo (1975)**, doctors must not take part in torture; **Sydney (1968)**, determination of death; **Oslo (1970)**, therapeutic abortion.

- **Organon §1:** the physician's high and only mission is to restore the sick to health. §2: the ideal cure is rapid, gentle and permanent.

### Duties and misconduct

- **Duties to the patient:** attend in an emergency, keep confidentiality, obtain consent, keep records, give proper prescriptions with the registration number, and refer when beyond one's competence.
- **Professional ("infamous") misconduct** includes: adultery or improper conduct with a patient; issuing false certificates; **advertising** or self-promotion; **fee-splitting (dichotomy)**; covering an unqualified person; disclosing a patient's secrets without cause; performing an illegal abortion or sex determination; practising another system of medicine without qualification.
- **Penalties:** warning or censure; temporary removal; **penal erasure** of the name from the register.
- **Cross-practice:** in *Poonam Verma v. Ashwin Patel* (1996) the Supreme Court held that a homoeopath prescribing allopathic drugs without qualification is **negligent per se**.

### Consent

- **Implied consent:** the patient's conduct shows agreement (e.g. coming to the clinic, holding out an arm). It covers only a general examination.
- **Expressed consent:** oral or written; written consent is needed for any procedure with risk.
- **Informed consent:** the patient is told the nature, risks, alternatives and consequences of refusal, in a language they understand.
- **Age:** a person aged **18 years or more** can give valid consent (IPC 87). For a child **under 12** or a person of unsound mind, the guardian consents (IPC 89). In an **emergency**, the doctor may act in good faith without consent (IPC 92). The BNS sections 25–30 carry these rules forward.
- **Blanket consent** ("any treatment") is not valid.
- A **medico-legal examination** of a victim needs that person's consent; an accused may be examined at police request without consent (BNSS 51, formerly CrPC 53).

### Professional negligence

- **Elements, the "4 Ds":** **D**uty of care, **D**ereliction (breach) of that duty, **D**irect causation, **D**amage.
- **Civil negligence** brings compensation; **criminal negligence** needs gross negligence or recklessness (IPC 304A, now **BNS 106**).
- **Bolam test:** a doctor is not negligent if a responsible body of fellow practitioners would have acted the same way.
- ***Jacob Mathew v. State of Punjab* (2005):** guidelines that protect doctors from frivolous criminal prosecution.

- ***Res ipsa loquitur*** ("the thing speaks for itself"): the facts prove negligence, e.g. a swab left in the abdomen or the wrong limb operated on.
- **Contributory negligence:** the patient's own carelessness (e.g. not following advice) reduces the doctor's liability.
- **Novus actus interveniens:** a new, unrelated act breaks the chain of causation.
- **Consumer protection:** *Indian Medical Association v. V.P. Shantha* (1995) brought paid medical service under consumer law. Under the **Consumer Protection Act, 2019**, claims go to the **District Commission (up to ₹50 lakh)**, **State Commission (₹50 lakh to ₹2 crore)** or **National Commission (above ₹2 crore)**.

### Confidentiality and records

- **Professional secrecy:** the doctor must not disclose what the patient tells them.
- **Privileged communication:** disclosure is justified when the law or the public interest requires it, e.g. notifiable diseases, a court order, a patient's infection endangering a spouse, or a danger to others (e.g. an epileptic bus driver).
- **Records:** keep in-patient records for **at least 3 years**; keep **medico-legal records until the case is closed**. A patient (or their legal heir) may ask for copies, which should be given within **72 hours**.
- The **Charter of Patients' Rights (NHRC, 2019)** lists rights such as information, records, emergency care, informed consent, confidentiality, a second opinion, non-discrimination, and choice of where to buy medicines and tests.

### Legal procedure and evidence

- **Inquest:** an inquiry into an unnatural death. **Police inquest** is the commonest in India; **magistrate inquest** is mandatory for deaths in custody, dowry deaths within 7 years of marriage and exhumation.
- **Medico-legal autopsy:** ordered by the police or magistrate; relatives' consent is not needed. A **clinical (academic) autopsy** needs the relatives' consent.
- **Dying declaration:** a statement by a dying person about the cause of death; best recorded by a magistrate, but a doctor may record it if death is imminent. The doctor certifies that the person is **conscious and of sound mind (compos mentis)**. No oath is given. A **dying deposition** is recorded by a magistrate in the presence of the accused, with cross-examination, and carries more weight.
- **Witness examination order:** examination-in-chief → **cross-examination** → re-examination. **Leading questions** are allowed only in cross-examination.
- **Perjury:** giving false evidence under oath (IPC 191, now BNS 227).
- **Summons and conduct money:** a doctor must obey a court summons; in a criminal case it cannot be refused for lack of fees.

### Other Acts a homoeopath must know

| Act | Key fact |
|---|---|
| Medical Termination of Pregnancy Act, 1971 (amended 2021) | Up to 20 weeks on one registered (allopathic) practitioner's opinion; 20–24 weeks on two, for special categories; beyond 24 weeks by a medical board for major foetal anomalies |
| PC-PNDT Act, 1994 (amended 2003) | Bans sex determination before or after conception |
| POCSO Act, 2012 | Age of child below 18; **mandatory reporting** of child sexual abuse by any doctor |
| Mental Healthcare Act, 2017 | A suicide attempt is presumed to be due to severe stress; it is no longer punished |
| Transplantation of Human Organs and Tissues Act, 1994 | Recognises brain-stem death; bans commercial organ trade |
| Drugs and Cosmetics Act, 1940 and Rules | Homoeopathic medicines are regulated here; **Schedule M-I** sets good manufacturing practice for homoeopathic medicines |
| Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954 | Bans advertising remedies that claim to cure listed diseases |

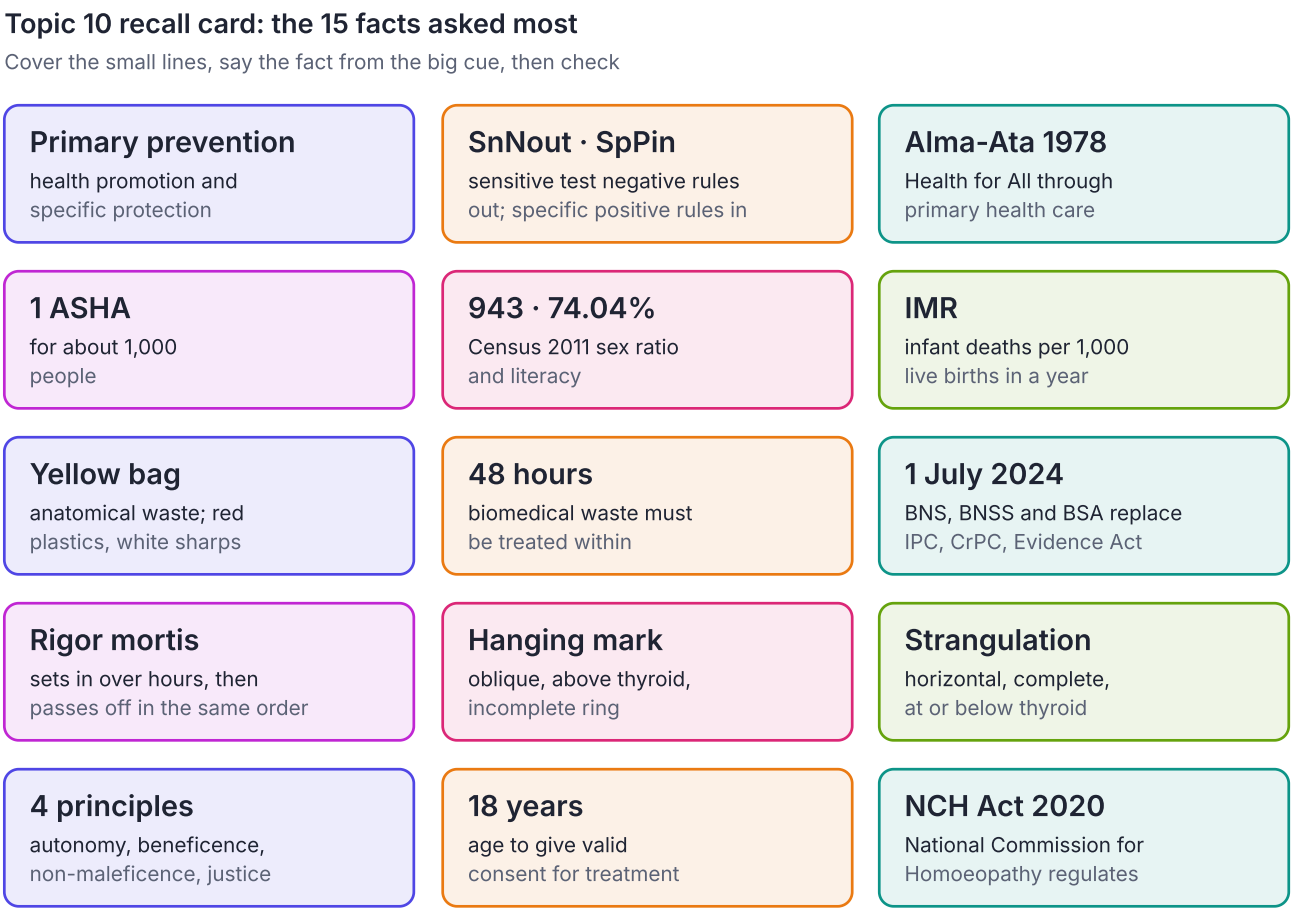
**Memory hook:** the four principles are "**A** **B**ig **N**ice **J**ob": **A**utonomy, **B**eneficence, **N**on-maleficence, **J**ustice. Valid consent for treatment needs a patient of **18 or over**, informed and free to refuse.

## Rapid revision

1. WHO definition of health: 1948; physical, mental and social well-being.
2. Epidemiological triad: **agent, host, environment**; web of causation for non-communicable disease.
3. Five modes of intervention: health promotion, specific protection, early diagnosis and treatment, disability limitation, rehabilitation.
4. Immunisation = **primary prevention (specific protection)**; screening = **secondary**; physiotherapy = **tertiary**.
5. **P = I × D**.

6. Case-control gives **odds ratio**; cohort gives **relative risk** and incidence.
7. Sensitivity = TP/(TP+FN); specificity = TN/(TN+FP); PPV rises with prevalence.
8. Cold chain **+2 to +8 °C**; OPV most heat-sensitive; Hep B and pentavalent must not freeze; shake test for freezing.
9. Genus epidemicus: Organon §100–102; Belladonna in scarlet fever (1799).
10. RBSK screens 0–18 years for 4 Ds.
11. Sub-centre 5,000; PHC 30,000; CHC 1,20,000 (plain areas).
12. IMR per **1,000 live births**; MMR per **1,00,000 live births**.
13. Replacement level: **NRR = 1**, TFR about 2.1.
14. Census 2011 sex ratio 943; literacy 74.04%.
15. Normal curve: ±1 SD 68%, ±2 SD 95%, ±3 SD 99.7%.
16. Chi-square for proportions; t-test for two small-sample means.
17. Residual chlorine **0.5 mg/L after 1 hour**; Horrock's apparatus; bleaching powder 33% chlorine.
18. Slow sand filter: **Schmutzdecke** (vital layer).
19. Fluoride above 1.5 mg/L causes fluorosis; Nalgonda technique.
20. BMW: **yellow** anatomical/soiled; **red** recyclable plastics; **white** sharps; **blue** glass and metal implants; treat within 48 hours.
21. Rigor follows **Nysten's rule**; first sign of putrefaction is green **right iliac fossa**.
22. Casper's dictum 1 : 2 : 8 (air : water : earth).
23. Cherry-red hypostasis: **CO and cyanide**.
24. Cadaveric spasm cannot be faked: weeds in drowning, weapon in suicide.
25. Hanging: oblique, high, incomplete mark; strangulation: horizontal, low, complete.
26. Hyoid fracture commonest in **throttling**; diatom test for drowning.
27. Datura: dry, red, blind, mad, hot; arsenic: Mees' lines, rice-water stools; lead: Burtonian line.
28. Elapids neurotoxic; vipers haemotoxic; **20-minute whole blood clotting test**.
29. IPC 302 → **BNS 103**; IPC 320 → **BNS 116**; CrPC 174 → **BNSS 194**; Evidence Act 32 → **BSA 26**.
30. NCH Act 2020 replaced the HCC Act 1973; three autonomous boards.
31. Negligence: duty, dereliction, direct causation, damage; *res ipsa loquitur*; Bolam test.
32. *Poonam Verma v. Ashwin Patel* (1996): cross-practice is negligence per se.
33. Valid consent from **18 years**; guardian for a child under 12; emergency without consent in good faith.
34. Leading questions only in **cross-examination**; dying declaration has no oath.
35. Consumer Protection Act 2019: District up to ₹50 lakh, State to ₹2 crore, National above.

## One-glance recall card




For the last ten minutes before the exam: cover the small lines of each tile and say the fact from the big cue.

## 20 practice MCQs

Cover the answer key, attempt all 20 in about 15 minutes, then check.

1. Specific protection belongs to which level of prevention? A. Primordial  B. Primary  C. Secondary  D. Tertiary
2. Prevalence equals: A. Incidence ÷ duration  B. Incidence × duration  C. Incidence + duration  D. Incidence − deaths
3. The odds ratio is calculated in a: A. Cohort study  B. Case-control study  C. Cross-sectional study  D. Field trial
4. Positive predictive value of a test increases when: A. Prevalence rises  B. Prevalence falls  C. Sensitivity falls  D. Sample size falls

5. Which vaccine is most sensitive to heat? A. Hepatitis B  B. BCG  C. OPV  D. Td
6. Hahnemann used Belladonna as a prophylactic against: A. Cholera  B. Scarlet fever  C. Smallpox  D. Typhus
7. Maternal mortality ratio is expressed per: A. 1,000 live births  B. 1,000 women 15–49  C. 1,00,000 live births  D. 1,00,000 population
8. India is in which stage of the demographic cycle? A. High stationary  B. Early expanding  C. Late expanding  D. Low stationary
9. The best measure of central tendency for skewed data is the: A. Mean  B. Median  C. Mode  D. Range
10. After chlorination, the free residual chlorine at 1 hour should be: A. 0.1 mg/L  B. 0.2 mg/L  C. 0.5 mg/L  D. 2 mg/L
11. Discarded glass vials go into the: A. Yellow bag  B. Red bag  C. White container  D. Blue cardboard box
12. The first external sign of putrefaction appears over the: A. Face  B. Right iliac fossa  C. Left iliac fossa  D. Back
13. Cherry-red hypostasis is seen in poisoning by: A. Arsenic  B. Carbon monoxide  C. Opium  D. Organophosphate
14. A ligature mark that is oblique, above the thyroid cartilage and incomplete suggests: A. Throttling  B. Hanging  C. Ligature strangulation  D. Smothering
15. Mees' lines on the nails are seen in chronic poisoning by: A. Lead  B. Mercury  C. Arsenic  D. Copper
16. The 20-minute whole blood clotting test is used in: A. Cobra bite  B. Krait bite  C. Viper bite  D. Scorpion sting
17. The punishment for murder is now under: A. BNS 101  B. BNS 103  C. BNS 116  D. BNSS 194
18. The National Commission for Homoeopathy Act was passed in: A. 1973  B. 2014  C. 2019  D. 2020
19. "The thing speaks for itself" refers to: A. Novus actus interveniens  B. Res ipsa loquitur  C. Bolam test  D. Contributory negligence
20. Leading questions may be asked during: A. Examination-in-chief  B. Cross-examination  C. Re-examination  D. None of these

**Answer key:** 1 B · 2 B · 3 B · 4 A · 5 C · 6 B · 7 C · 8 C · 9 B · 10 C · 11 D · 12 B · 13 B · 14 B · 15 C · 16 C · 17 B · 18 D · 19 B · 20 B

**Why the tricky ones:** Q4, PPV depends on how common the disease is. Q8, India's birth rate is falling but still above its death rate. Q11, glass and metal implants go in the blue box, not the red bag. Q16, viper venom stops blood clotting; elapid venom is neurotoxic.